Antenna Digital Twin: Models and Parameters Technical Reference

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Abstract

This document details the models used across the Antenna Digital Twin backend and the parameters each module consumes and produces. For each domain—Environment, Learning, Measurement, Mechanical, Optimization, and Validation—we describe *what* the model does and *why* it exists in the system.

Contents

1 System Overview

The Antenna Digital Twin backend is organized into six main domains. The following TikZ flow illustrates how these domains interact.

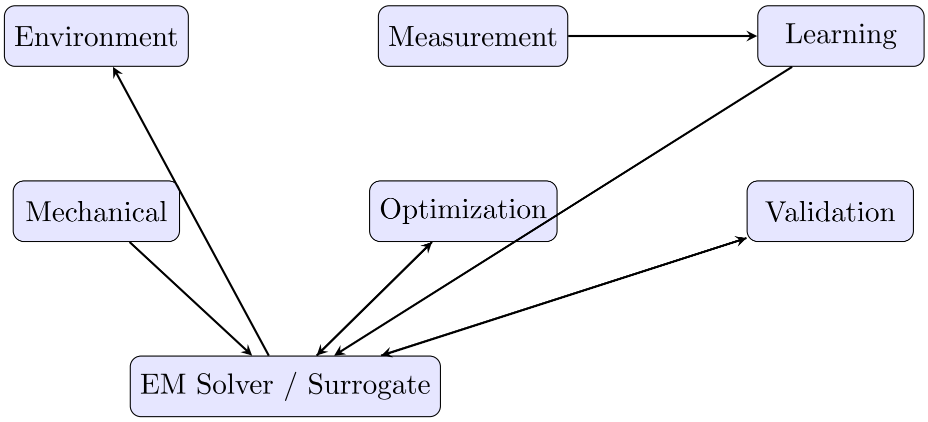

Figure 1: High-level flow of the six backend domains relative to EM simulation and surrogate inference.

2 Environment

2.1 What

The Environment domain models how the antenna behaves in its deployment setting: propagation loss, coverage, proximity effects (hand, housing), and orientation.

2.2 Why

Antenna performance depends on the operating environment. Indoor vs. outdoor, distance from obstacles, and mounting orientation all affect effective gain and S11. These models enable the digital twin to predict performance under real-world conditions, not just in free space.

2.3 Models and Parameters

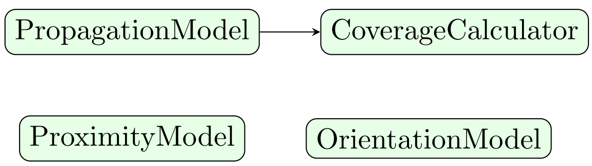

Figure 2: Environment model dependencies. Coverage uses Propagation; Proximity and Orientation act on pattern/parameters.

3 Measurement

3.1 What

The Measurement domain ingests data from VNA and chamber files, aligns it with known antenna parameters, and validates correctness.

Model	Input / Output	Reason
PropagationModel	distance (m), frequency (Hz), environment → path_loss (dB)	*What*: Free-space path loss $20\log_{10}(4\pi d/\lambda)$ with optional indoor (+20 dB) or outdoor (+10 dB). *Why*: Predicts received power at distance and environment.
CoverageCalculator	pattern, distance, frequency, env, threshold_gain → coverage_prob, effective_gain	*What*: PropagationModel + max gain; checks if effective gain > threshold. *Why*: "Will the link work at distance d?"
ProximityModel	parameters, obstacle_distance, permittivity → S11 degradation factor	*What*: $\exp(-d/0.02) \cdot (\varepsilon_r - 1) \cdot 0.1$ for d<0.1 m; no effect >10 cm. *Why*: Hand/housing detunes resonance.
OrientationModel	pattern, theta/phi rotation (deg) → RadiationPattern	*What*: Placeholder for rotation. *Why*: Mounting orientation affects apparent gain.

Table 1: Environment models, parameters, and rationale.

3.2 Why

Physical measurements (S11, gain, efficiency) are the ground truth for calibrating the digital twin. Without reliable ingestion and alignment, learning and validation cannot use real-world data.

3.3 Models and Parameters

Component	Input / Output	Reason
MeasurementIngestionService	file_path, file_type, antenna_id, params, metadata → MeasurementData	*What*: Parses VNA/Chamber, validates, aligns. *Why*: Central entry point.
VNAParser, ChamberParser	file path → parsed dict (freq, s11, gain, eff)	*What*: Format-specific parsers. *Why*: Different equipment formats.
ParameterAligner	measurement_data, known_params → AntennaParameters	*What*: 1% tolerance check; aligned params. *Why*: Correct design association.
MeasurementValidator	parsed_data → valid, errors	*What*: Validates freq range, S11 bounds, fields. *Why*: Reject malformed data.

Table 2: Measurement components, parameters, and rationale.

4 Learning

4.1 What

The Learning domain calibrates the surrogate using measurements, performs Bayesian updates, detects drift, and triggers retraining when needed.

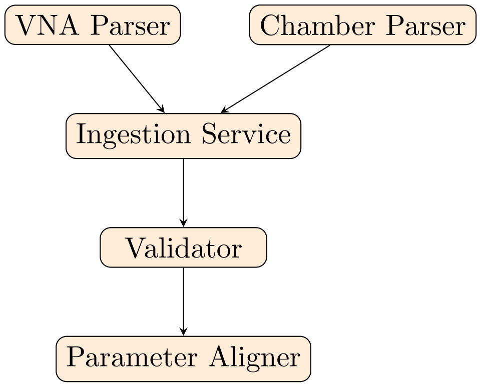

Figure 3: Measurement ingestion pipeline.

4.2 Why

The surrogate is trained on EM simulations, which can differ from physical measurements due to manufacturing tolerances, substrate variation, and environment. Learning keeps the twin aligned with reality.

4.3 Models and Parameters

Model	**Input / Output**	**Reason**
CalibrationService	antenna_id, measurement, db → discrepancy, bayesian_update, confidence	*What*: Compares pred vs. meas (S11, gain, eff), Bayesian update, confidence. *Why*: Aligns twin with physical instance.
BayesianUpdater	model_pred, measurement, prior_var → updated_s11, variance, weight	*What*: $0.7 \cdot \text{meas} + 0.3 \cdot \text{pred}$; reduces variance. *Why*: Bayesian fusion.
DriftDetector	predictions, measurements, threshold=0.1 → drift_detected, mean_error	*What*: Rel. error pred vs. meas S11; drift if mean > 0.1. *Why*: Detects model mismatch.
AutomatedBayesianUpdater	antenna_id, measurement_id → calibration, drift_detection	*What*: Orchestrates calibration + drift on new measurement. *Why*: Single entry point.

Table 3: Learning models, parameters, and rationale.

5 Mechanical

5.1 What

The Mechanical domain models deformation of the antenna geometry due to bending, mounting stress, and thermal expansion.

5.2 Why

Flexible substrates (e.g., FR-4 on curved surfaces), mounting stress, and temperature changes alter physical dimensions and thus resonance and gain. Modeling these effects makes the twin useful for conformal and stressed designs.

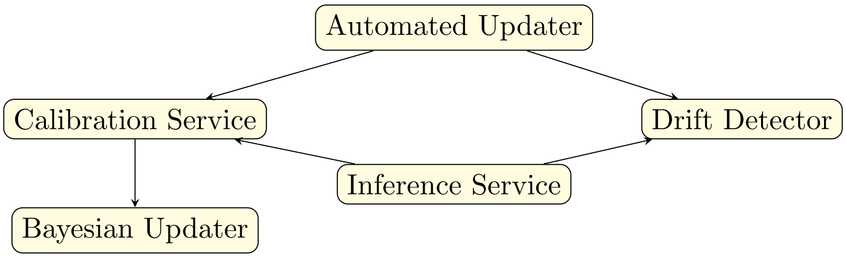

Figure 4: Learning flow: Automated updater triggers calibration and drift detection; both use inference predictions.

5.3 Models and Parameters

Model	Input / Output	Reason
BendingModel	params, bending_radius, axis → AntennaGeometry	*What*: Strain $\varepsilon = h/(2R)$; $L' = L(1+\varepsilon)$, $W' = W(1-\nu\varepsilon)$. $E = 2.4\,\mathrm{GPa}$, $\nu = 0.22$. *Why*: Bending changes L, W.
StressModel	params, stress_x, stress_y (Pa) → AntennaGeometry	*What*: $\varepsilon_x = \sigma_x/E$, $\varepsilon_y = \sigma_y/E$ + Poisson. *Why*: Mounting stress stretches patch.
ThermalExpansionModel	params, temp (°C), $T_{\mathrm{ref}} = 25$ → AntennaGeometry	*What*: $1 + \alpha\Delta T$, $\alpha = 16 \times 10^{-6}\,\mathrm{K}^{-1}$. *Why*: Temp shifts resonance.
GeometryDeformer	params, bending, stress, temp → AntennaParameters	*What*: bending → stress → thermal sequence. *Why*: Single interface.

Table 4: Mechanical models, parameters, and rationale.

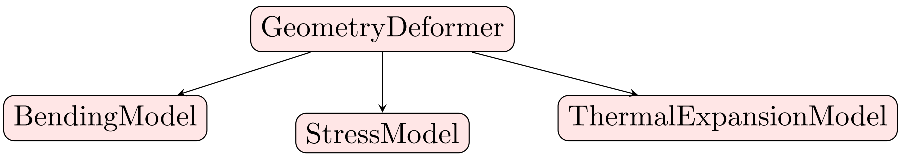

Figure 5: Mechanical deformation pipeline: GeometryDeformer chains bending, stress, and thermal.

6 Optimization

6.1 What

The Optimization domain tunes antenna geometry to meet targets: single-value S11, full S11 spectrum (UCE-style), and what-if analysis.

6.2 Why

Engineers need to find geometries that achieve desired S11, gain, or efficiency. Single-target optimization (e.g., S11 = $-10\,\mathrm{dB}$) is common; spectrum matching supports multi-band and measured-target workflows. What-if analysis explores sensitivity without running full optimization.

6.3 Models and Parameters

Model	Input / Output	Reason
GeometryOptimizer.optimize_s11	initial_params, target_s11 (dB), bounds → AntennaParameters	*What*: L-BFGS-B on surrogate; min |s11_min − target|. *Why*: Gradient-based single target.
optimize_s11_spectrum	initial, target_freq/s11_db, optimizer, quantile, n_samples, elite_frac, n_iter → params, loss_history	*What*: UCE spectrum loss + L-BFGS-B or CEM. *Why*: Match full S11 curve.
spectrum_loss	pred/target freq and S11_db, quantile → scalar	*What*: Piecewise loss (squared/mixed L1), quantile threshold. *Why*: Robust; UCE design.
cross_entropy_optimize	func, bounds, n_samples, elite_frac, n_iter → best_x, score, history	*What*: CEM uniform sampling; elite updates dist. *Why*: Derivative-free.
WhatIfAnalyzer	base_params, variation → Dict[str, SurrogatePrediction]	*What*: Surrogate for base + varied params. *Why*: Sensitivity exploration.

Table 5: Optimization components, parameters, and rationale.

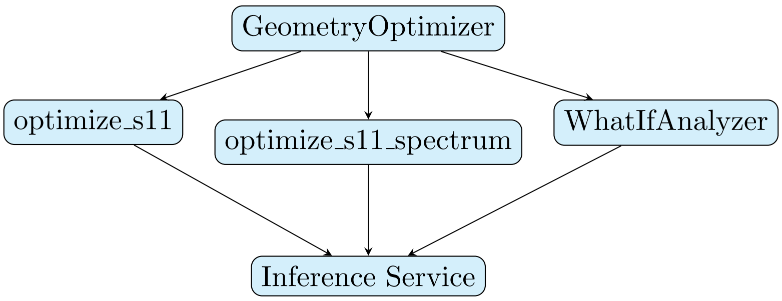

Figure 6: Optimization flows. All use Inference Service (surrogate) for fast evaluations.

7 Validation

7.1 What

The Validation domain compares surrogate predictions to EM ground truth and tracks KPIs over time.

7.2 Why

Surrogates must stay within acceptable error bounds. Validation quantifies MAE, RMSE, R^2, and max error so engineers know when to retrain or trust predictions.

Model	Input / Output	Reason
EMSurrogateValidator	em_results, surrogate_preds, metric → MAE, MRE, RMSE, max_err, R2	*What*: EM vs. surrogate comparison. *Why*: Surrogate accuracy.
KPITracker	kpi_name, value, metadata → JSON file	*What*: Logs KPIs with timestamp. *Why*: Track metrics over time.

Table 6: Validation components, parameters, and rationale.

7.3 Models and Parameters

8 Core Parameters and Schemas

8.1 AntennaParameters

The central input to EM simulation, surrogate inference, mechanical deformation, and optimization.

Field	Type	What & Why
geometry.length	float (m)	Patch length L; resonance.
geometry.width	float (m)	Width W; bandwidth.
geometry.height	float (m)	Substrate h; Q, bandwidth.
geometry.feed_x, feed_y	float (m)	Feed; impedance match.
substrate.substrate_type	FR4, RO4003, RO4350	Material; ε_r, tan δ.
substrate.relative_permittivity	float	ε_r; resonance.
substrate.loss_tangent	float	tan δ; losses.
frequency_band	2.4GHz, 3.5GHz	Band; default freq range.
frequency_range	(f_min, f_max) Hz	Sim/pred frequency span.

Table 7: AntennaParameters schema.

8.2 Surrogate Model Parameters

The ensemble combines GP and NN; features are the 7 antenna parameters (length, width, height, feed_x, feed_y, ε_r, tan δ). Target metrics: `s11_min`, `gain`, `efficiency`.

Model	Key Parameters	What & Why
GaussianProcessSurrogate	input_dim=7, Matern($\nu = 2.5$)	GP with uncertainty; confidence intervals.
NeuralSurrogate	input_dim=7, hidden [64,128,64], dropout 0.2	MLP; fast inference, scales to more data.
EnsemblePredictor	weighted_avg (0.6 GP, 0.4 NN)	Blends GP+NN; GP for uncertainty.

Table 8: Surrogate model parameters.

9 Configuration (Settings)

Key configuration values that affect behavior:

10 Conclusion

This report documented the models and parameters used in the Antenna Digital Twin backend. Each domain—Environment, Measurement, Learning, Mechanical, Optimization, and

Setting	Default	What & Why
ML_MODEL_DIR	models/	Surrogate pickles.
ML_DEVICE	cpu	PyTorch device; cuda for GPU.
EM_SOLVER_TIMEOUT	3600 s	Max sim time per run.
EM_SOLVER_MAX_WORKERS	4	Parallel EM for batch training.
POSTGRES_*	localhost, 5432, antenna_twin	Database for instances, measurements.

Table 9: Key configuration parameters.

Validation—has a clear role: Environment models deployment conditions; Measurement ingests and aligns physical data; Learning calibrates and detects drift; Mechanical models deformation; Optimization tunes geometry; Validation ensures surrogate accuracy. The core parameters (AntennaParameters, surrogate features, and configuration) are defined to support 2.4 GHz and 3.5 GHz microstrip patch antenna twinning with minimal coupling between modules.